Requirements Analysis for “Alzheimer's Dementia Recognition through Spontaneous Speech” software development

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Functional Requirements

- **Speech capture and analysis**
  - recording well-defined speech samples
  - validation of quality and quantity
  - extraction of sound and language features

- **Testing for cognitive impairment**
  - being able to input novel speech samples into the model
  - calculating multi-dimensional data (at least most if not all useful features)
  - generate feedback in case of potential issues
  - be time-effective in evaluation

- **Neuroimage compatibility**
  - proper integration of MNE toolkit
  - near-universal possibility of comparison with neurological data

- **Data management**
  - maintain functionality in case of missing or corrupt data
  - allow a sufficiently diverse source of data
  - track evolution

- **Classification and prediction**
  - evaluation of probability of mental impairment
  - if possible, prediction of likelihood of developing such conditions

- **Report generation**
  - generation of an overall report in an easily readable format

- **Ethical considerations**
  - verify the legal framework when obtaining data used when training the model

Non-functional Requirements

- **Performance**
  - adequate response time
  - clear and effective scoring
  - ability to process larger (likely GBs) volumes of data
  - efficient database retrieval

- **Reliability**
  - low to very low likelihood of collapse (system and model)
  - fault-tolerance (database, fallback modes, replication)
  - error-handling (failures, validation, partial data)

- **Security and data integrity**
  - caution against potential undesirable infection
  - caution against data poisoning

- **Privacy**
  - follow all the ethical and legal standard in place
  - allow the possibility of removing data